		Period of waveform in option B = 4 cm × 0.5 ms cm^{-1} = 2 ms
23	**A**	Magnetic force provides centripetal force → $Bqv = \frac{mv^2}{r}$ → $Bq = \frac{mv}{r}$ → $r = \frac{mv}{Bq}$ $Bq = \frac{mv}{r} = m\omega = \frac{2\pi m}{T}$ → $T = \frac{2\pi m}{Bq}$
24	**C**	The magnetic force acting on the proton is **upwards**